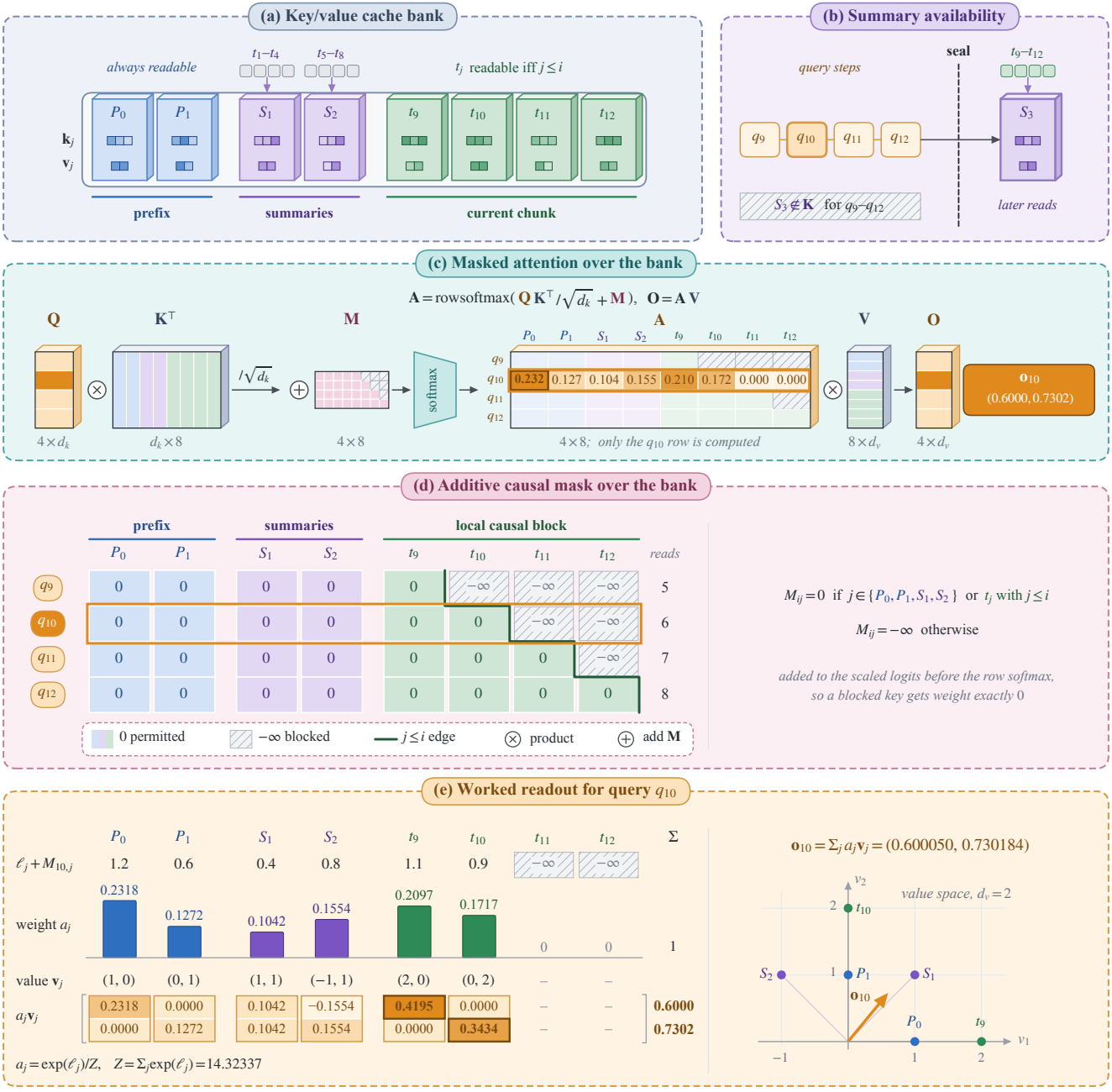

(a) Key/value cache bank
always readable
t_1–t_4
t_5–t_8
t_j readable iff $j \le i$
$\mathbf{k}_j$
$\mathbf{v}_j$
P_0
P_1
S_1
S_2
t_9
t_{10}
t_{11}
t_{12}
prefix
summaries
current chunk
(b) Summary availability
query steps
seal
t_9–t_{12}
q_9
q_{10}
q_{11}
q_{12}
S_3
$S_3 \notin \mathbf{K}$ for q_9–q_{12}
later reads
(c) Masked attention over the bank
$\mathbf{A} = \mathrm{rowsoftmax}(\mathbf{Q}\mathbf{K}^\top/\sqrt{d_k} + \mathbf{M}),\ \mathbf{O} = \mathbf{A}\mathbf{V}$
Q
$\mathbf{K}^\top$
M
A
V
O
$/\sqrt{d_k}$
softmax
0.232
0.127
0.104
0.155
0.210
0.172
0.000
0.000
$\mathbf{o}_{10}$
(0.6000, 0.7302)
$4\times d_k$
$d_k\times 8$
4×8
4×8; only the q_{10} row is computed
$8\times d_v$
$4\times d_v$
(d) Additive causal mask over the bank
prefix
summaries
local causal block
reads
$M_{ij} = 0$ if $j \in \{P_0, P_1, S_1, S_2\}$ or t_j with $j \le i$
$M_{ij} = -\infty$ otherwise
added to the scaled logits before the row softmax, so a blocked key gets weight exactly 0
0 permitted
$-\infty$ blocked
$j \le i$ edge
product
add M
(e) Worked readout for query q_{10}
$\ell_j + M_{10,j}$
1.2
0.6
0.4
0.8
1.1
0.9
$-\infty$
$-\infty$
Σ
weight a_j
0.2318
0.1272
0.1042
0.1554
0.2097
0.1717
0
0
1
value $\mathbf{v}_j$
(1, 0)
(0, 1)
(1, 1)
(−1, 1)
(2, 0)
(0, 2)
$a_j\mathbf{v}_j$
0.2318
0.0000
0.0000
0.1272
0.1042
0.1042
−0.1554
0.1554
0.4195
0.0000
0.0000
0.3434
0.6000
0.7302
$a_j = \exp(\ell_j)/Z$, $Z = \Sigma_j \exp(\ell_j) = 14.32337$
$\mathbf{o}_{10} = \Sigma_j a_j \mathbf{v}_j = (0.600050, 0.730184)$
value space, $d_v = 2$
v_2
v_1
t_{10}
P_1
S_2
S_1
$\mathbf{o}_{10}$
P_0
t_9